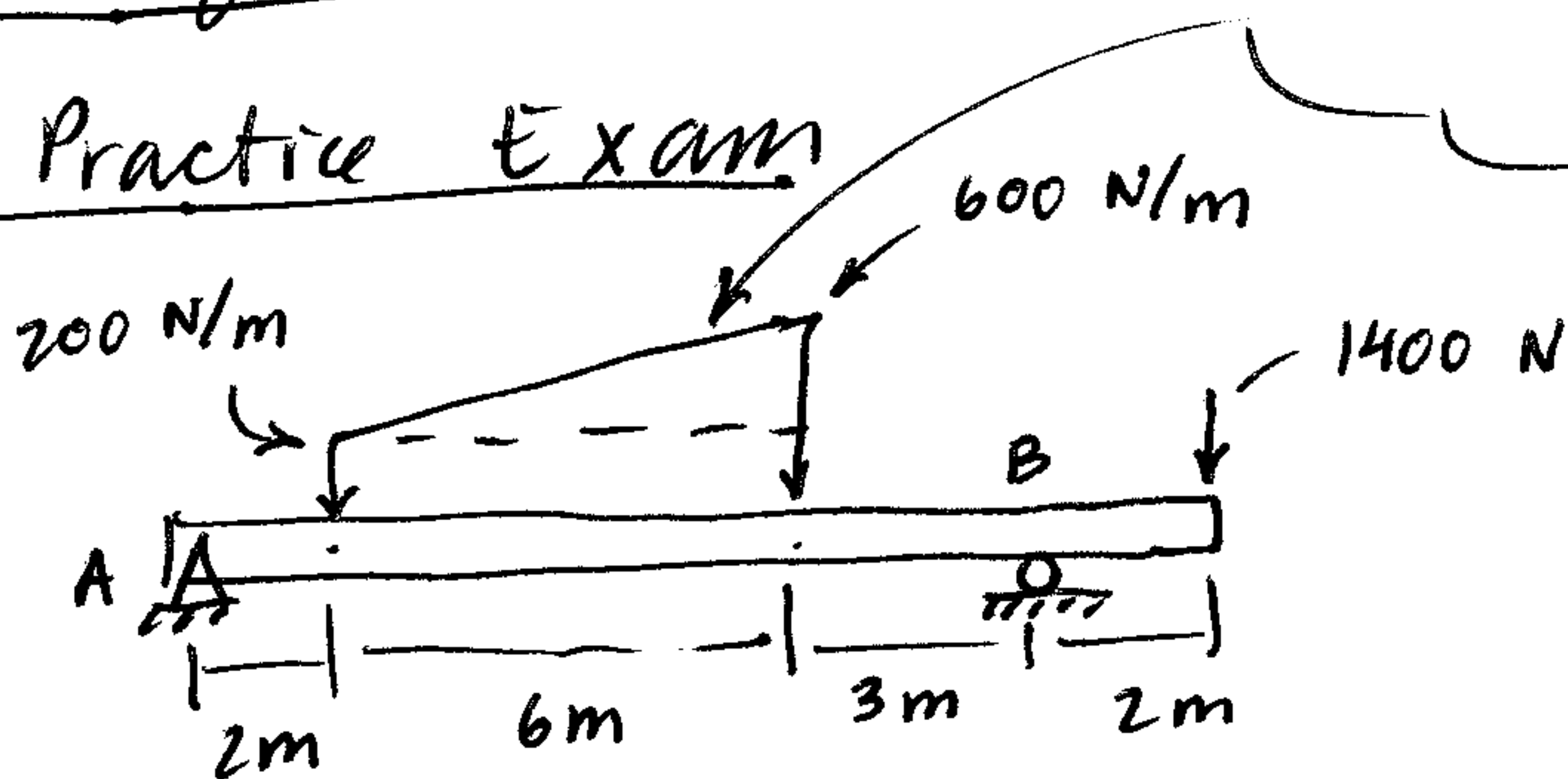


V&M Egn's

①

2022 Practice Exam



$$\text{Slope} = \frac{600 - 200}{6} = \frac{400}{6} = 66.67$$

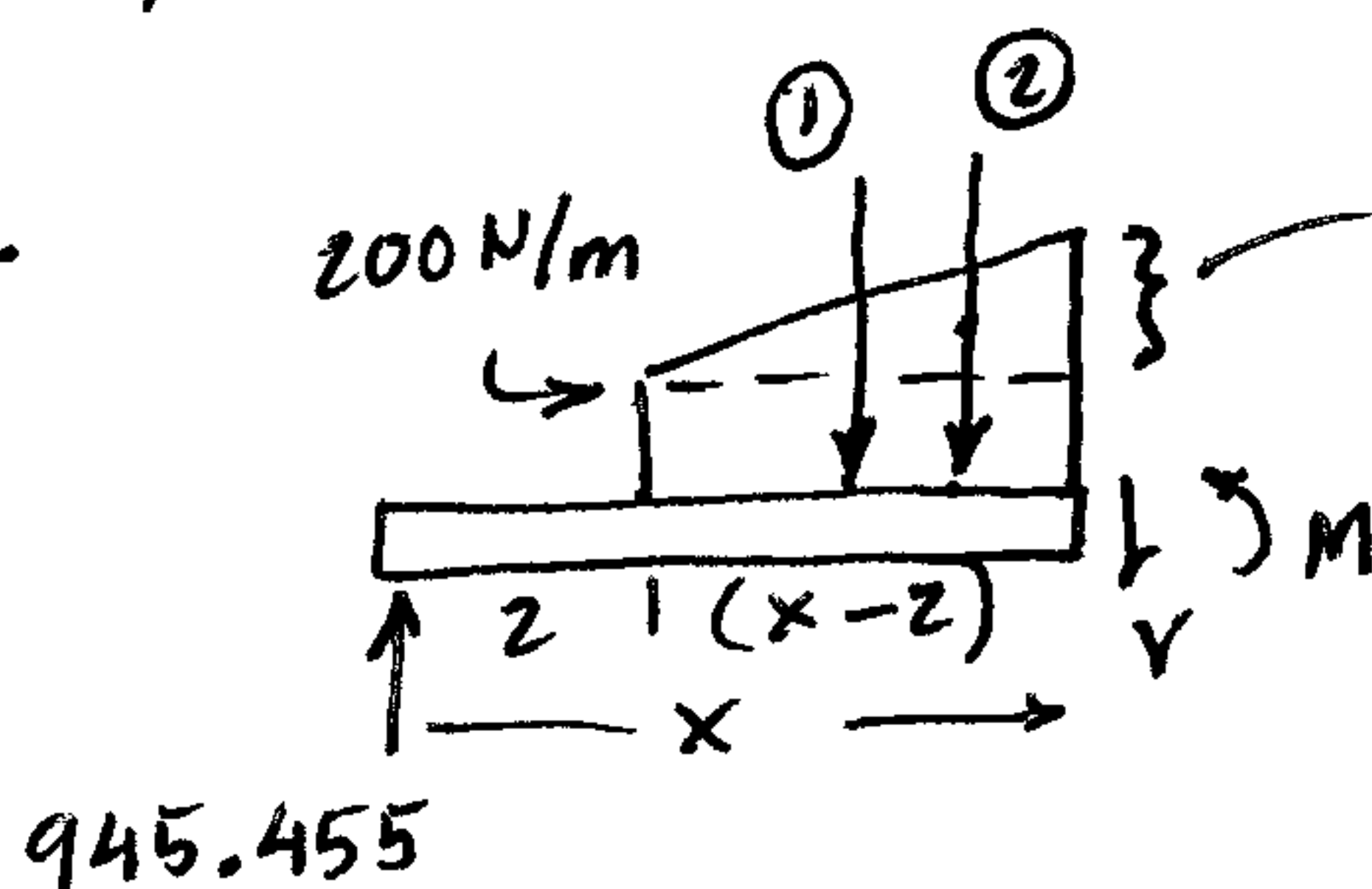
For the interval $2 \leq x \leq 8$, the exam gives:

$$V = -33.33(x^2 + 2x - 36.364)$$

Can we verify this? Obtain this equation?

$$2 \leq x \leq 8$$

FBD



$$\text{slope} \cdot (x-2) = 66.67(x-2)$$

$$\textcircled{1} = 200(x-2)$$

$$\textcircled{2} = \frac{1}{2}bh$$

$$= \frac{1}{2}(x-2) \cdot 66.67(x-2)$$

$$= 33.33(x-2)^2$$

$$+ \uparrow 945.455 - 200(x-2) - 33.33(x-2)^2 - V = 0$$

$$[-200x + 400]$$

$$-33.33(x^2 - 4x + 4)$$

$$= [-33.33x^2 + 133.33x - 133.33]$$

Gathering like terms:

$$-33.33x^2$$

$$-200x + 133.33x = -66.67x$$

$$945.455 + 400 - 133.33 = 1212.1$$

We get:

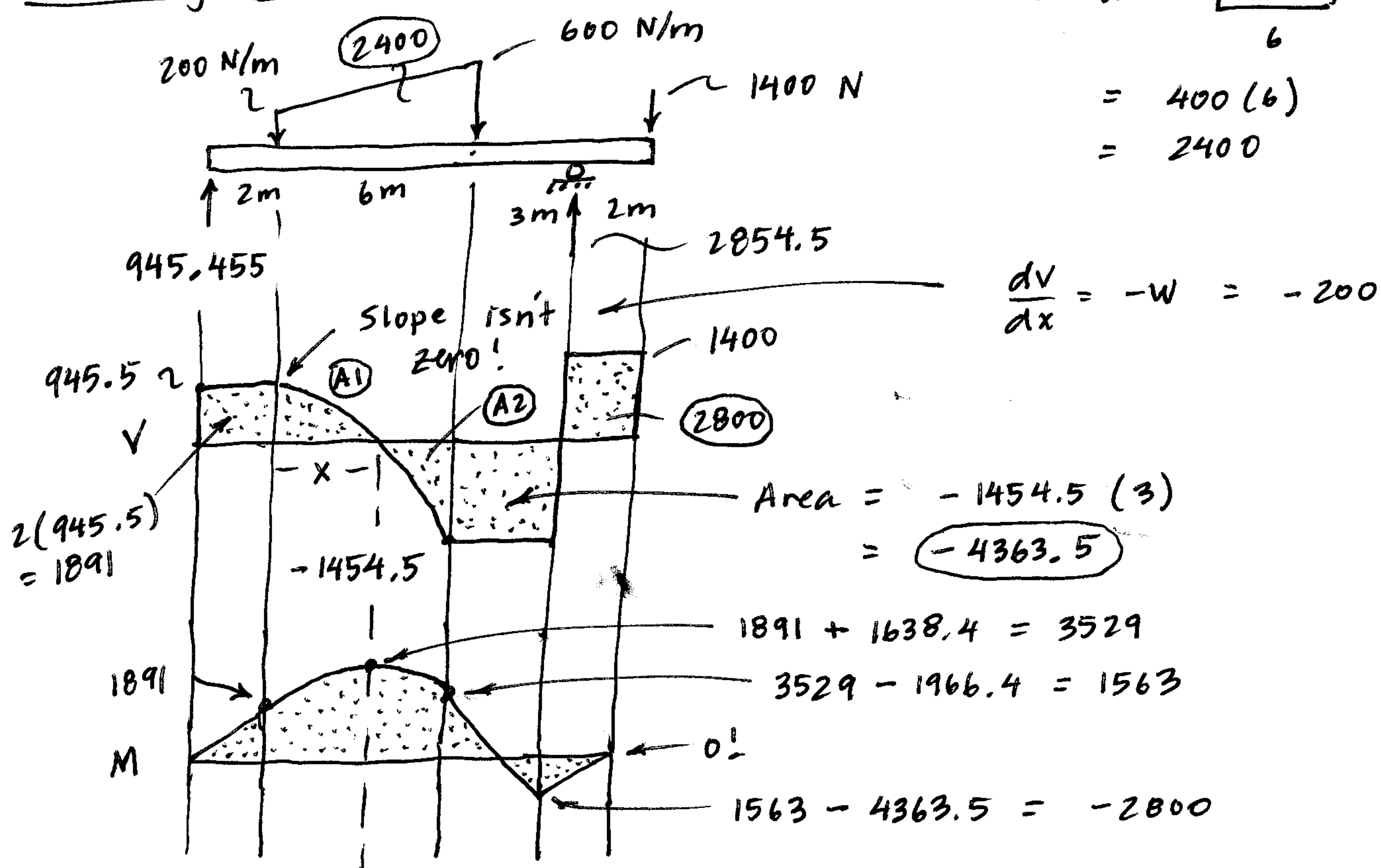
$$V = -33.33x^2 - 66.67x + 1212.1$$

or...

$$V = -33.33[x^2 + 2x - 36.36]$$

Drawing the V & M Graphs

Area: $\frac{200 + 600}{2} \times 6$ (2)
 $= 400(6)$
 $= 2400$



We obtained for $2 \leq x \leq 8$:

$$V = -33.33(x^2 + 2x - 36.36)$$

To find x , use `zeros()` function (TI-89 or nSpire)

$$\text{zeros}(-33.33(x^2 + 2x - 36.36), x)$$

$$= 5.112 = x$$

To find areas A1 and A2, integrate w/ TI-89 or nSpire.

$$A1 = \int (-33.33(x^2 + 2x - 36.36), x, 2, 5.112)$$

$$A1 = 1638.4$$

$$A2 = \int (-33.33(x^2 + 2x - 36.36), x, 5.112, 8)$$

$$A2 = -1966.4$$